

Boqi Chen

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EDUCATION

Dept. of Computer Science, ETH Zurich

Doctor of Philosophy; ETH AI Center Doctoral Fellowship (acceptance rate: 1%)

Zurich, Switzerland

Jun 2027 (Expected)

Dept. of Information Technology and Electrical Engineering, ETH Zurich

Master of Science; GPA: 5.69/6

Zurich, Switzerland

Mar 2023

Dept. of Electrical Engineering, Xi'an Jiaotong University

Bachelor of Engineering; GPA: 88/100

Xi'an, China

Jun 2020

RESEARCH EXPERIENCE

ETH AI Center

Doctoral Assistant

Zurich, Switzerland

Jun 2023 – Present, Full-time

- **Vision Foundation Model (VFM):** Researched on automatic data curation for VFM training. Curated a high-quality balanced dataset from 350 million public images using hierarchical clustering. Deployed large-scale distributed pretraining of Vision Transformer (ViT) using DINOv2 on 32×H200 clusters.
- **Diffusion Models:** Researched on cross-modality domain generalization in image segmentation based on invariant causal mechanisms. Proposed a diffusion-based augmentation technique using text-guided image editing. Implemented finetuning of *Stable Diffusion* and *ControlNet* using *diffusers*.

IBM Research Europe

Machine Learning Intern

Zurich, Switzerland

Sep 2021 – Nov 2022, Full-time

- **Weakly-Supervised Learning:** Researched on algorithms for gigapixel-sized image classification. Proposed an end-to-end trainable multiple instance learning method using differentiable perturbed optimizers, optimizing model architecture for enhanced computational efficiency.
- **Generative Adversarial Networks (GANs):** Proposed a generative image replay algorithm utilizing neural style transfer and GANs for continual unsupervised domain adaptation in image segmentation.

SKILLS

Programming: Python, C/C++ (intermediate), MATLAB (intermediate), LaTeX

AI/ML Frameworks: PyTorch, PyTorch Lightning, Scikit-learn, Hugging Face, OpenCV

Development Tools: Git, Linux, Docker

PUBLICATIONS

[1] **B. Chen***, et al., "Revisiting Automatic Data Curation for Vision Foundation Models in Digital Pathology," in *MICCAI*, 2025.

[2] S. Andani*, **B. Chen***, et al., "Histopathology-based Protein Multiplex Generation using Deep Learning," *Nature Machine Intelligence* (Accepted).

[3] **B. Chen**, et al., "Generalizable Single-Source Cross-modality Medical Image Segmentation via Invariant Causal Mechanisms," in *WACV*, 2025.

[4] **B. Chen***, K. Thandiackal*, P. Pati, and O. Goksel, "Generative Appearance Replay for Continual Unsupervised Domain Adaptation," *Medical Image Analysis*, vol. 89, 102924, Elsevier, 2023.

[5] K. Thandiackal*, **B. Chen***, et al., "Differentiable Zooming for Multiple Instance Learning on Whole-Slide Images," in *ECCV*, 2022.